

Signal Processing

Exercise Sheet 5

SoSe26

Part 1 – Sampling Theorem and Reconstruction

Question 1 – Sampling Frequency and Aliasing

The continuous-time signals $x_1(t)$ and $x_2(t)$ are given by

$$x_1(t) = \sin(2\pi 100t) \quad (3)$$

and

$$x_2(t) = \cos(4000\pi t), \quad (4)$$

respectively. For case of simplicity, we assume both signals to be T_s -spaced sampled once again.

In practice, the effect aliasing needs to be addressed. It is for this reason that signals are often sampled at rates much higher than twice the highest frequency component contained in the signal of interest. Determine, if aliasing might turn out to be in issue in case the sampling time for $x_1(t)$ and $x_2(t)$ is chosen to be

$$T_1 = \frac{1}{400} \text{ s}$$

and

$$T_2 = \frac{1}{1500} \text{ s},$$

respectively.

Optional: Verify your results by sketching the frequency spectra of the discrete-time signals $x_1(n)$ and $x_2(n)$.

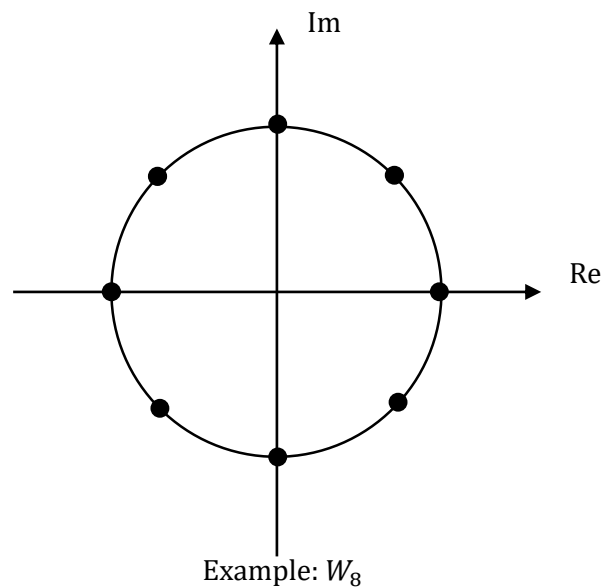
Part 2 – Discrete Fourier Transform (DFT)

The DFT allows for the numerical calculation of the spectrum of a signal. Keep in mind that the number of acquired signal values N corresponds to the same number of calculated frequency components M .

Discrete Fourier Transform

$$X_{\text{DFT}}[m] = \sum_{n=0}^{N-1} x[n] e^{-jn2\pi m/N} = \sum_{n=0}^{N-1} x[n] \cdot W_N^{nm},$$

where $W_N = e^{-\frac{j2\pi}{N}}$ is the rotation phasor and $m = 0, \dots, N-1$.



Inverse Discrete Fourier Transform

$$x[n] = \frac{1}{N} \sum_{m=0}^{N-1} X_{\text{DFT}}[m] \cdot W_N^{-nm},$$

where $m = 0, \dots, N-1$.

Question 1

The sequences $x_0[n]$, $x_1[n]$ and $x_2[n]$ are given by

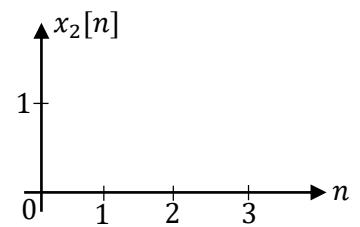
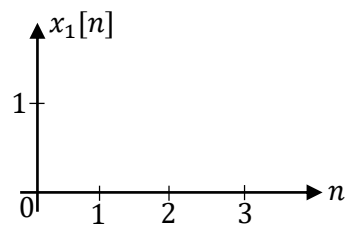
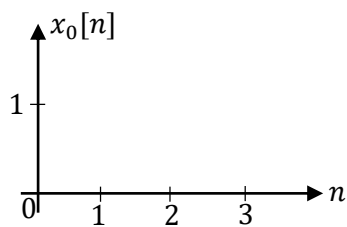
$$x_0[n] = \delta[n],$$

$$x_1[n] = \delta[n - 1],$$

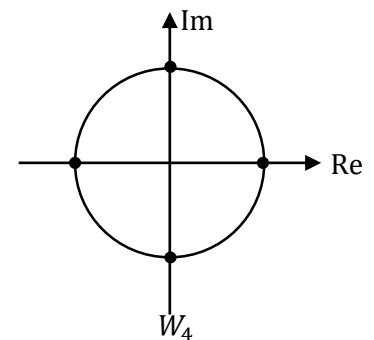
$$x_2[n] = \delta[n - 2],$$

where $n = 0, \dots, N - 1$.

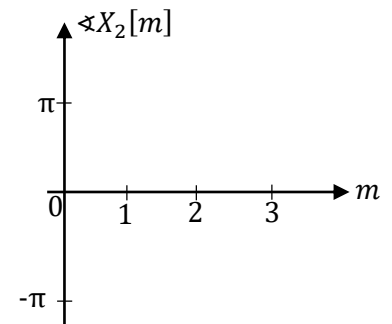
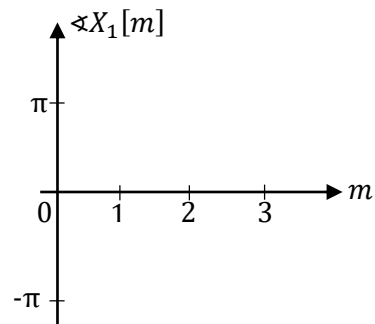
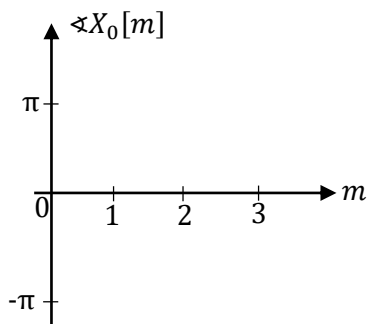
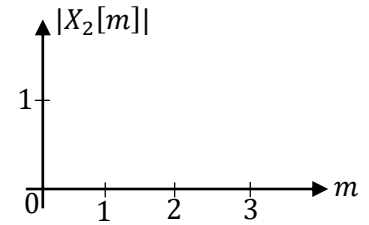
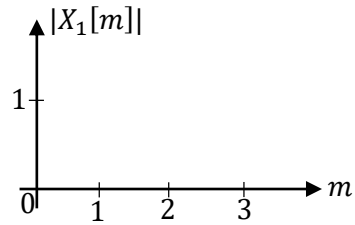
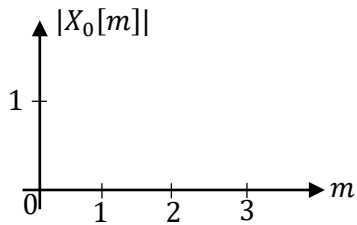
a) Sketch the sequences for $N = 4$.



b) Calculate the DFT_4 for each sequence.



c) Sketch the magnitude and phase of $X_0[m]$, $X_1[m]$ and $X_2[m]$, respectively.



Question 2

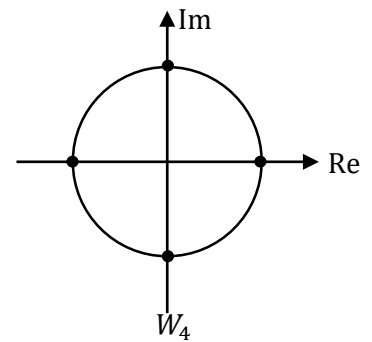
The impulse response $h[n]$ and the input signal $x[n]$ of an LTI system are given by

$$h[n] = [2, 0, -1, 1]$$

and

$$x[n] = [3, 4, 2, 1].$$

a) Determine the DFT_4 of $h[n]$ and $x[n]$.



b) Determine the DFT of the output signal $y[n]$ of the system.